

Treat the following exercises :

Exercise 1 (3 pts)

Observe the following XML document:

```
<?xml version="1.0"?>
<note date=3 janvier>
  <to>Bob</to>
  <from>Alice</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend !</body>
</note>
<note date= '5 janvier' <!--this is another note -->>
  <to>Bob</to>
  <from>Bob
  <body>No problem & see you soon</body>
</note>
<note/>
```

a) Is this document well-formed ?
b) If it is not, correct the errors.

Exercise 2 (4 pts)

In this exercise we work on an XML document, Course.xml which contains information about computer courses as indicated in Annex. Write the result of the transformation applied to the following document Course .xml.

Electronic Commerce
Class room: Amphi A
Modules : PHP and MySQL, XML
Registered (note):
I235678 (16)
I784451
F569834

Database
Class room: Painlevé
Modules : SQL, Optimisation
Registered (note):
A675432 (14)
B455978
B568709 (12)

Exercise 3 (5 pts)

Here is an extract of a file containing the list of winners of the Booker prize (list of books with their author and year of receiving the award).

```
<?xml version="1.0"?>
<booker>
  <award>
    <author>Kingsley Amis</author>
    <title>The Old Devils</title>
    <year>1986</year>
  </award>
  <award>
    [...]
  </award>
  [...]
</booker>
```

Find the expressions XPath that returns the following information (we do not suppose that only the initial context is the root element of name booker):

- the author's sixth book in the list
- the title of the book that won in 2000
- the name of the author of the book entitled "Possession"
- the title of the books where "JM Coetzee" is the author
- the name of every author whose book has won since 1995

Exercise 4 (4 pts)

We wish represent an address book in XML. For each entry in the book, we want to keep the following information:

- last name and first name of the person
- address (possibly outside of Lebanon)
- phone number (possibly many)
- email address (possibly many)
- birthday date

a) Write a DTD for the address book.

b) Write a valid xml file for that DTD, including at least two entries and showing all the possibilities of the DTD (i.e. all the possible values for attributes and all optional elements and attributes).

Exercise 5 (4 pts)

Represent graphically as a tree the following xml file:

```
<?xml version= "1.0" ?>
<memo>
<to>head of departement</to>
<from> author</from>
<cc>
<staffperson>
<rank>professor</rank>
<name>jane jones</name>
<office>101 maclief</office>
</staffperson>
</cc>
<subject>student scores</subject>
<body>
Congratulations everyone! our students averaged over 90 percent
</body>
</memo>
```